CS & IT ENGINEERING



Algorithms

Divide & Conquer



Lecture No.- 08

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Recap of Previous Lecture



Topic

Topic

Matrix Multiplication.

Master's method

Topics to be Covered



- Topic
- Topic
- Topic

Master's method

Long Integer Multiplication

(Q.1) $T(n) = 4T(n/2) + n^2 \longrightarrow \Theta(n^2 \log n)$

(Q.2) $T(n) = 9T(n/3) + \log n \longrightarrow \underline{\Theta(n^2)}$

$$T(n) = 4T(n/2) + \log n$$

$\log_3 9 = 2$

$\log_2 4 = 2$

(Q.1) $T(n) = 4T(n/2) + n^2$

Soln:

$a = 4$
$b = 2$
$F(n) = n^2$

$\log_b a = \log_2 (4) = 2$

(ans): If $n^2 = O(n^{2-\varepsilon})$, $\varepsilon > 0$?

False

Ans 2 :- Is $n^2 = \Theta(n^2 * (\log n)^k)$

1) $k \geq 0$, For $k = 0$, $n^2 = \Theta(n^2 * (\log n)^0)$

$n^2 = \Theta(n^2)$ ✓

True

$$T(n) = \Theta\left(n^2 (\log n)^{0+1}\right) = \Theta(n^2 \log n)$$

(Q.2) $T(n) = 9T(n/3) + \log n$ ⟶

$a = 9$
$b = 3$
$f(n) = \log n$ } ✓

$\log_b a = \log_3 9 = \underline{2}$

Case 1: Is $\log n = O(n^{2-\varepsilon})$, some $\varepsilon > 0$?

$\varepsilon = 0.5, 0.1$ ✓

True

$$T(n) = O(n^2)$$

7.2.1

(Q.3) $T(n) = T(n/3) + n$

a) $\Theta(n^2)$

b) $\Theta(n)$

c) $\Theta(\sqrt{n})$

d) $\Theta(\log n)$

$$T(n) = T(n/3) + n$$

$a = 1$
$b = 3$
$F(n) = n$

} ✓ valid

$\log_3 1 = 0$

Case 1 :- Is $n = O(n^{0-\varepsilon})$, some $\varepsilon > 0$?

$$n = O(n^{-\varepsilon}),$$

False

Case 2 :- Is $n = O(n^0 * (\log n)^k)$

$n = O((\log n)^k)$

1) $k \geq 0$? X

2) $k = -1$, X

} Fails

Case 3:

Is $n = \Omega(n^{0+\varepsilon})$, some $\varepsilon > 0$.

$n = \Omega(n^{0.5})$ ✓

and

$a * f(n/b) \leq \delta * f(n)$

$1 * n/3 \leq \delta * n$

True

$1/3 \leq \delta$, $\delta < 1$

$T(n) = \Theta(n)$

(Q) $T(n) = 4T(n/2) + \log n$ 86%

A) $\Theta(n)$

B) $\Theta(\sqrt{n})$

C) ~~$\Theta(n^2)$~~ ✓

D) $\Theta(\log n)$

$a = 4$

$b = 2$

$f(n) = \log n$

$\log_2 4 = 2$

Case 1 : Is $\log n = O(n^{2-\varepsilon})$, some $\varepsilon > 0$?

$\log n = O(n)$ ✓

True

$\boxed{T(n) = O(n^2)}$

(9) $T(n) = 2^n T(n/4) + n$ → (inadmissable case)

$T(n) = aT(n/b) + f(n)$

$a \geq 1$, $b > 1$ } Constants

$f(n) = +ve$

$a = 2^n$ → invalid

$b = 4$

$f(n) = n$

☆ Advance variant of master's method.

[Special Case]

$$T(n) = 2T(\sqrt{n}) + \log_2 n$$

$$T(n) = 2T(\sqrt{n}) + \log_2 n$$

$$T(n) = 2T(n^{1/2}) + \log_2 n \quad \text{——— (1)}$$

Let $n = 2^k$ { change of variable method }

$n = 2^k \rightarrow k = \log_2 n$

$n^{1/2} = 2^{k/2}$

$T(n) = 2T(\sqrt{n}) + \log_2 n$ ——— (1) (given)

Put $n = 2^k$ in (1)

$T(2^k) = 2T(2^{k/2}) + \log_2(2^k)$

$T(2^k) = 2T(2^{k/2}) + k$ ——— (2)

Let $T(2^k) = P(k)$

then $T(2^{k/2}) = P(k/2)$

$P(k) = 2P(k/2) + k$ ——— (3)

0 1 0

$$P(k) = 2P(k/2) + k$$

$a = 2$

$b = 2$

$f(k) = k$

✓

$$\log_b a = \log_2 2 = 1$$

Case 1: Is $K = O(k^{1-\varepsilon})$, some $\varepsilon > 0$?

Fails.

Case2 : Is $f(n) = \Theta(n^{\log_b a} * (\log n)^k)$

Is $f(k) = \Theta(k^{\log_b a} * (\log k)^d)$

$k = \Theta(k^1 * (\log k)^d)$

a) $d \geq 0$? For $d = 0$, $k = \Theta(k)$ ✓

Yes

$P(k) = \Theta(k * (\log k)^{0+1})$

$P(k) = \Theta(k \log k)$

$P(k) = T(2^k) = O(k \log k)$

$T(n) = O(k \log k)$

$k = \log_2 n$

Ans :- $O(\log_2 n * \log(\log_2 n))$

$T(n) \rightarrow$

Final

561

$$T(n) = T(\sqrt{n}) + 5$$

A) $\Theta(\log n)$

B) $\Theta(n \log n)$

C) $\Theta(n)$

D) $\Theta(\log(\log n))$

$$T(n) = T(\sqrt{n}) + 5$$

(Change of variable mtd)

Let $n = 2^k$

$$\sqrt{n} = n^{1/2} = 2^{k/2}$$

$k = \log_2 n$

$$T(2^k) = T(2^{k/2}) + 5 \quad \text{—— (1)}$$

Let $T(2^k) = P(k)$

then $T(2^{k/2}) = P(k/2)$

$$T(2^k) = T(2^{k/2}) + 5$$

$$P(k) = P(k/2) + 5 \quad \text{---} \quad (2)$$

$a = 1$

$b = 2$

$f(k) = 5$ } Valid

↳ const ✓

$$\log_b a = \log_2 1 = 0$$

Case 1 :- $S = O(k^{0-\varepsilon})$, some $\varepsilon > 0$?

Fails

Case 2 :- Is $S = O(k^0 * (\log k)^d)$

some $d \geq 0$?

$d = 0$, $S = O(1 * 1)$

$S = O(\text{const})$ ✓ True

$P(k) = O(k^0 (\log k)^{0+1}) = O(\log k)$

$$P(k) = \Theta(\log k)$$

$$T(2^k) = \Theta(\log k)$$

$$T(n) = \Theta(\log k)$$

But $k = \log_2 n$

Hence, $T(n) = \Theta(\log(\log n))$ ✓

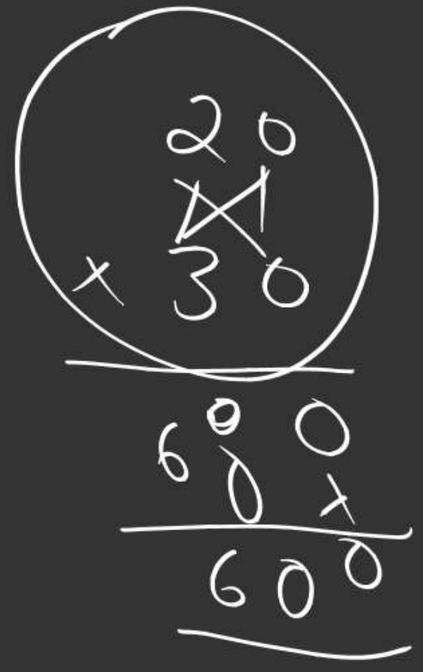
(Bonus) Long Integer Multiplication (LIM)

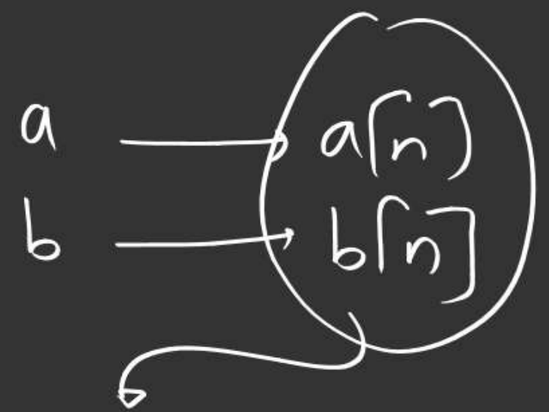
int a & b

a+b
a*b
} O(1)

3257139523410 ⟶ | 3 | 2 | 5 | 7 | 1 | 3 | . . . . . |

53795173298 25 ⟶ | 5 | 3 | 7 | 9 | 5 | 1 | . . . . . |



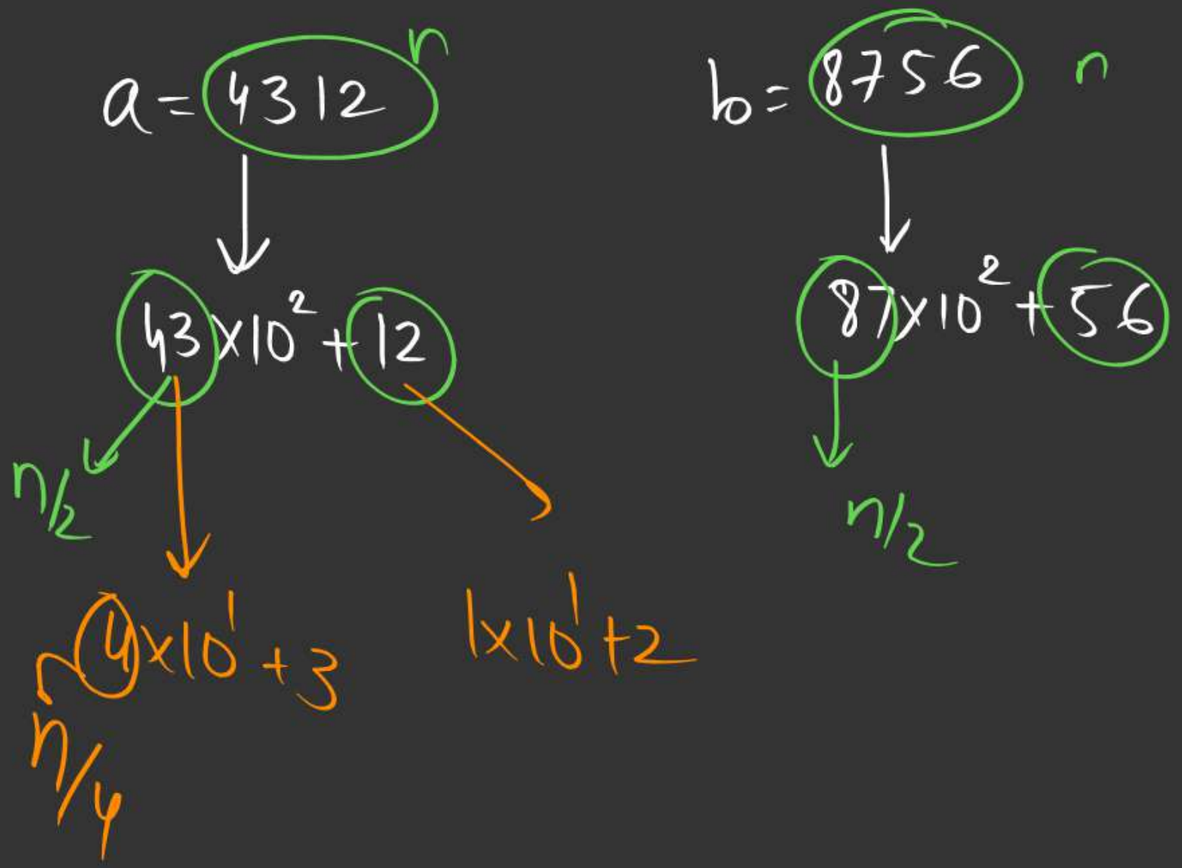


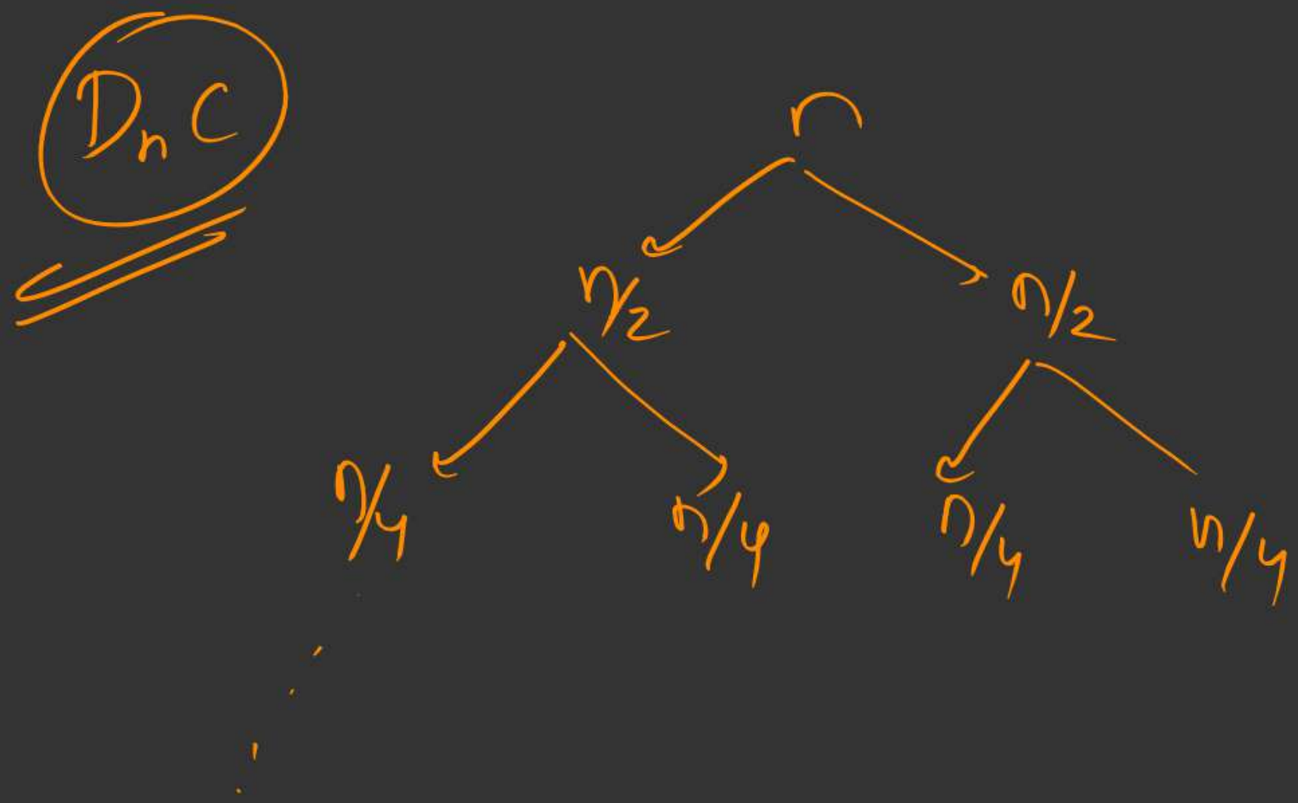
1) $a \pm b \longrightarrow O(n)$

2) $a * b \longrightarrow O(n^2)$

DnC Appz :

eg




D n C
n
n/2
n/2
n/4
n/4
n/4
n/4

Idea $a, b \longrightarrow n$ digit

Let $w, x, y, z \longrightarrow n/2$ digits

Let $m = \lfloor n/2 \rfloor$



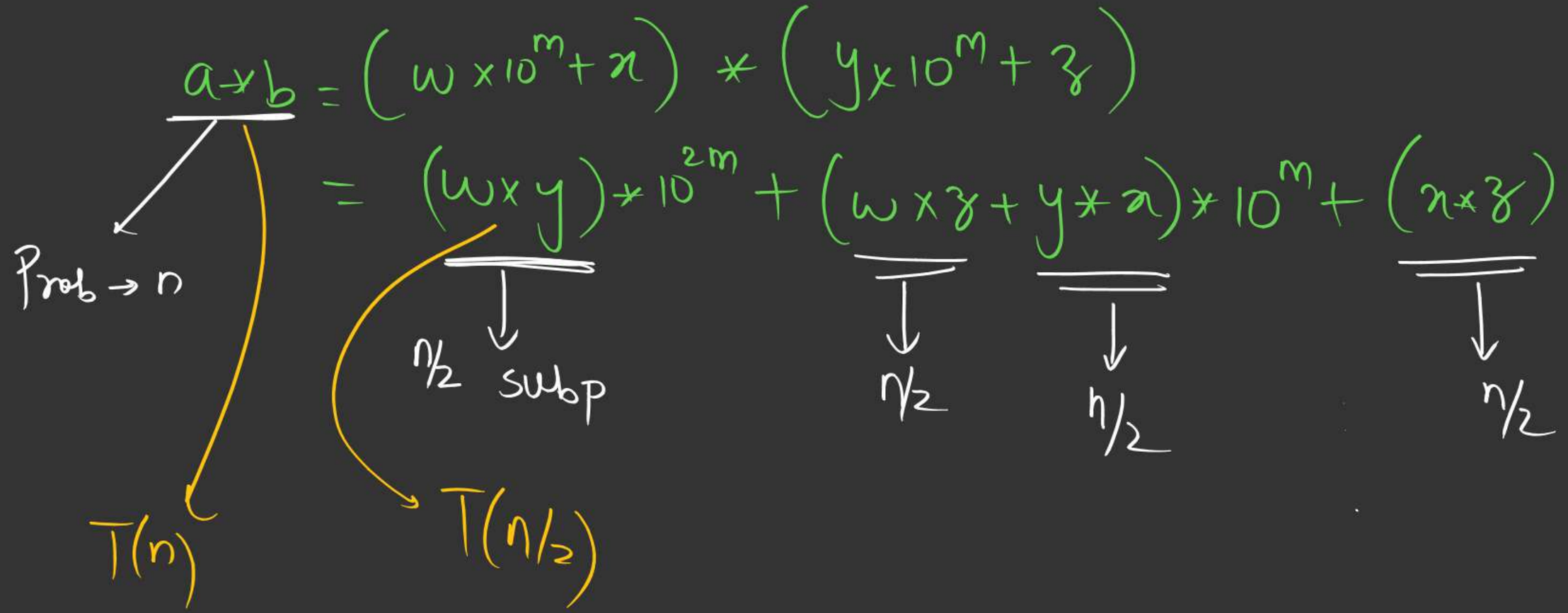
$$\begin{bmatrix} w = a/10^m, & x = a \% 10^m \\ y = b/10^m, & z = b \% 10^m \end{bmatrix}$$

$$\begin{bmatrix} a = w \times 10^m + x \\ b = y \times 10^m + z \end{bmatrix}$$

4312 → $43 \times 10^2 + 12$

$w = 4312/100$

$x = 4312 \% 100$


a*b = (w x 10^m + x) * (y x 10^m + z)
= (w x y) * 10^2m + (w x z + y * x) * 10^m + (x * z)
Prob → n
n/2
subp
n/2
n/2
n/2
T(n)
T(n/2)



Topic : Divide & Conquer

Continue:

a, b → n digit nos w, x, y, z

L\M m = [n/2] n/2 digits

$a \times b = (\underline{w \times y}) \times 10^{2m} + (\underline{w\,z} + \underline{x\,y}) \times 10^{m} + \underline{x\,y})$

n n/2 n/2 n/2 n/2

Sub problems



Topic : Divide & Conquer

Let T(n) represent time complexity of multiplying two long integer of size n digit each

$(ab)_n = (w\ y)_{n/2} * 10^{2m} + (wz + xy) \times 10^m + (xz)_{n/2}$

Sub problem of size n/2



Topic : Divide & Conquer

$T(n) = 4T(n/2) + bn, n > 1$

$= C, n = 1$

Master mtd

Adding → $TC = \theta(n^2)$ → LIM → Recurrence

mm

$O(n^3) \rightarrow O(n^{2.81})$

$T(n) = aT(n/b) + f(n)$

$a = 4, b = 2, f(n) = bn$ → valid

Case I : is $f(n) = 0\ ({}_n\log b^a - \varepsilon)$, same $\varepsilon > 0$

$b_n = 0\ (n^{2-\varepsilon}), \varepsilon > 0$? → Yes

→ Satisfied



Topic : Divide & Conquer

$T(n) = \theta({}_n \log b^a) = \theta(n^2)$

~~Meet mule~~ mat mul

(1) Non DnC → TC = $\theta(n^3)$

(2) Simple DnC → TC = $\theta(n^3)$

(3) Stressen's DnC → TC = $\theta(n^{2.81})$

Similarly:

for LTM DnC → AK optimization



Topic : Divide & Conquer

For AK optimization

$T(n) = 3T(n/2) + bn, \quad n > 1$

$= C, \quad n = 1$

$a = 3, \; b = 2, \; f(n) = bn$

Case 1:

$bn = O(_n\log_2 3 - \varepsilon), \; \varepsilon > 0$

→ satisfied yes

$T(n) = \theta(n^{1.58})$

$n^3 \rightarrow n^{2.81}$

$n^2 \rightarrow n^{1.58}$

$\log_2 3$

$n^3 \longrightarrow n^{2.81}$

$n^2 \longrightarrow n^{1.58}$



Topic : Divide & Conquer

2-way Skplit

Discussion so for:-

(1) Non-DnC → TC = $\theta(n^2)$

(2) Simple-DnC → TC = $\theta(n^2)$

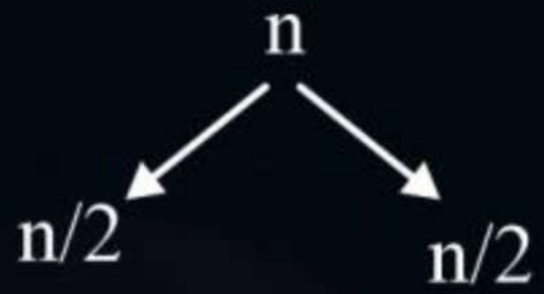
(3) AK Optimization → TC = $\theta(n^{1.58})$



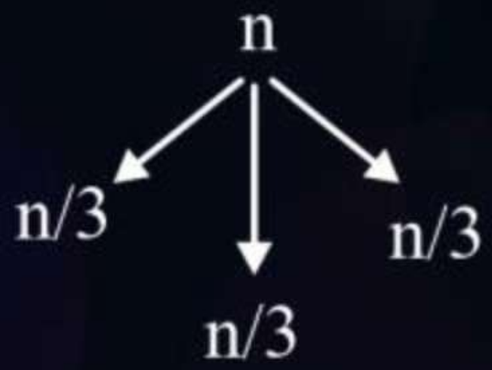
Topic : Toom & cook optimization

Idea:- (Bonus)

2-way split



3-way split



Topic : Recurrence after 3-way split

(1) Simple DnC $\Rightarrow$ $T(n) = 9T(n/3) + n$

$n^{\log_3^9} = n^2$

(2) AK optimization DnC $\Rightarrow$ $T(n) = 8T(n/3) + n$

$n^{\log_3^8} = n^{1.89}$

(3) T & C Optimization DnC $\Rightarrow$ $5T(n/3) + n$

$n^{1.46}$

$n^{\log_3 5}$



Topic : Summary to use in GATE

Summary to use in GATE

(1) 2-way split: DnC → 4, AK → 3, T & C → 3

(2) 3-way split: DnC → 9, AK → 8, T & C → 5

(3) 4-way split: DnC → 16, AK → 15, T & C → 7.

H.W validate pattern

Recurrence equation,

(1) DnC → $16T(n/4) + bn$, $n > 1$

(2) AK opti. → $15T(n/4) + bn$, $n > 1$

(3) T&C opt. → $7T(n/4) + bn$, $n > 1$



Topic : Summary to use in GATE

Generalised for k-way split

DnC → k^2 Ak → $(k^2 - 1)$ T&C → $(2k - 1)$

The generalized time complexity recurrences for k-way split:

(1) DnC → $T(n) = k^2\, T(n/k) + bn$

(2) AK opti. → $T(n) = (k^2 - 1)\, T(n/k) + bn$

(3) T&C opt.→ $T(n) = (2k - 1) T(n/k) + bn$





Topic : Divide & Conquer

#Q. The running time of an algorithm is represented by the following recurrence relation:

$$T(n) = \begin{cases} n & n \leq 3 \\ T\left(\frac{n}{3}\right) + cn & otherwise \end{cases}$$

Which of the following represents the time complexity of the algorithm?

H.W 1

A $\theta(n)$

B $\theta(n \log n)$

C $\theta(n^2)$

D $\theta(n^2 \log n)$



Topic : Divide & Conquer

#Q. QS on an unsorted list of size n, $(n/4)^{th}$ smallest element is selected as pivot each time, then the time complexity recurrence is ____.

Quick Sort

HW2



2 mins Summary

- Topic
- Topic
- Topic
- Topic
- Topic

Masters Method

Special Case:

D_nC ✓



Lim → normal

AK

TC



THANK - YOU